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The Transformer Explained: A Beginner's Guide to Attention-Based Neural Networks



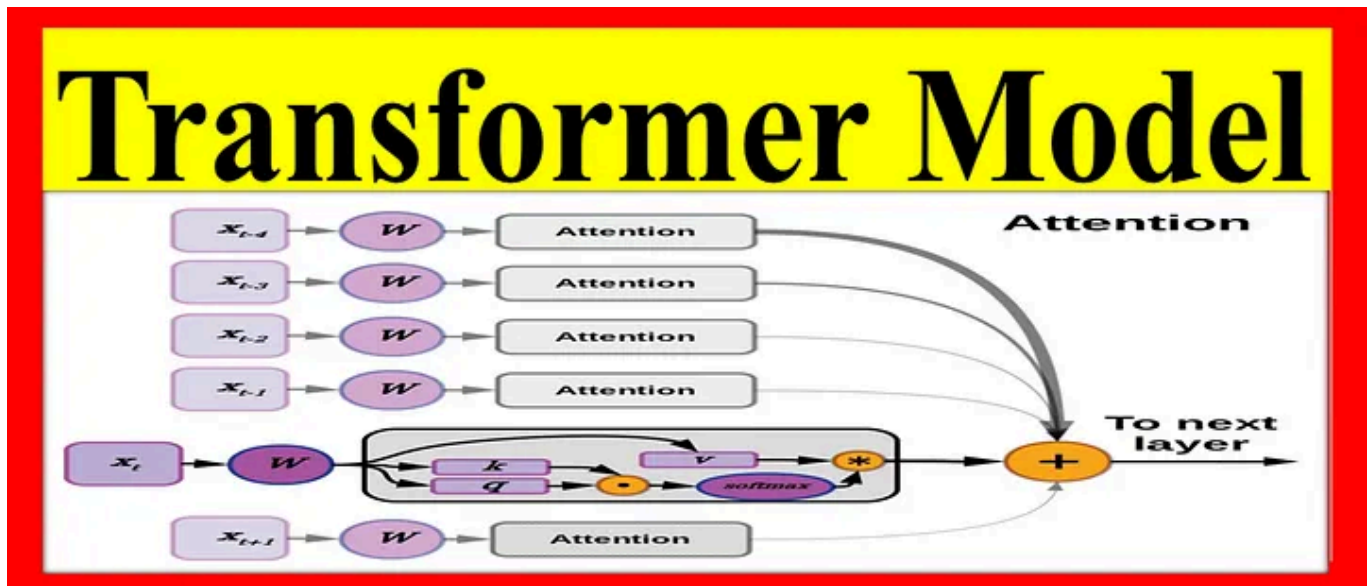
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The Transformer model has revolutionized natural language processing (NLP) by introducing the power of attention mechanisms to neural networks. **This groundbreaking architecture has become the backbone of many state-of-the-art NLP models, from machine translation to text generation.** In this blog post, we'll provide a comprehensive introduction to Transformer models, discussing the key concepts and components that make them so powerful. Let's get started!!

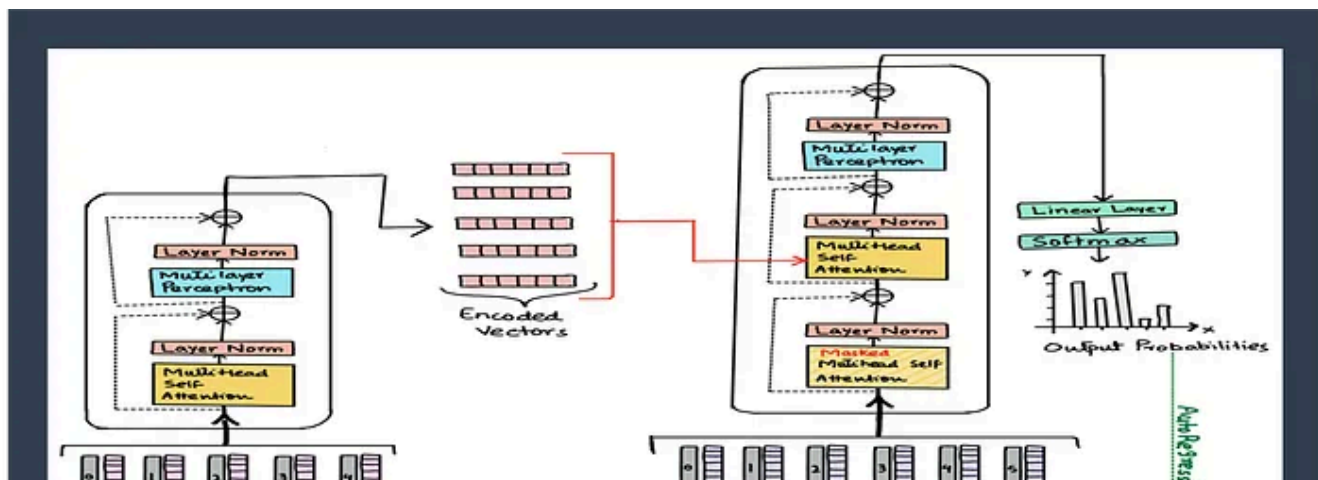


Transformer Model

What is a Transformer Model?

A Transformer is a popular and powerful deep-learning architecture used in natural language processing (NLP). It's particularly effective for tasks like language translation, text summarization, and question-answering.

At a high level, the Transformer is a neural network designed to process sequential data, such as text, by using a combination of self-attention mechanisms and feed-forward neural networks. The architecture is organized into “encoder” and “decoder” components.



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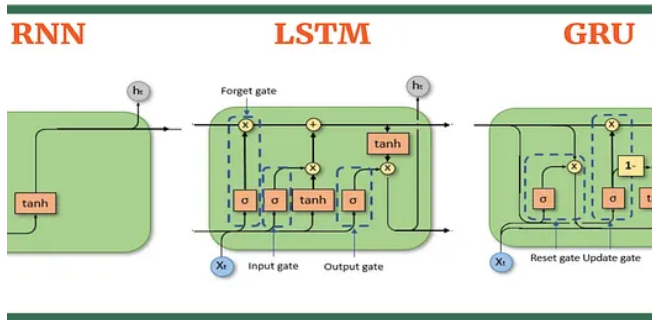
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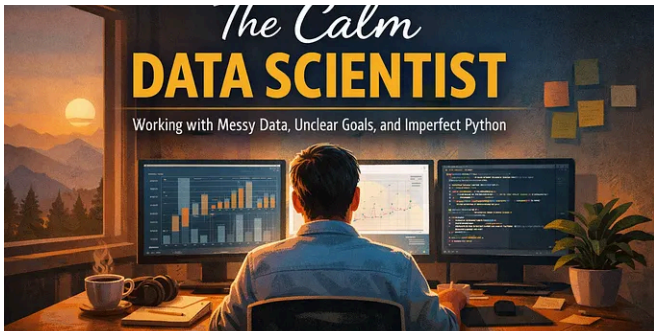
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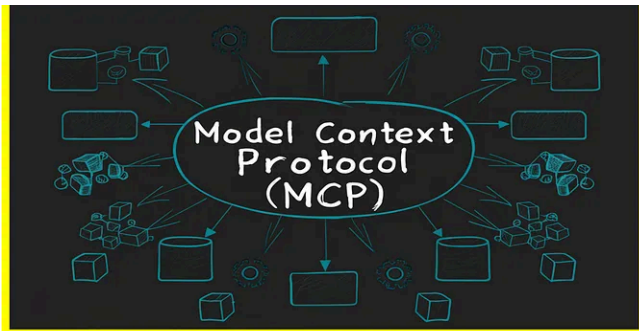


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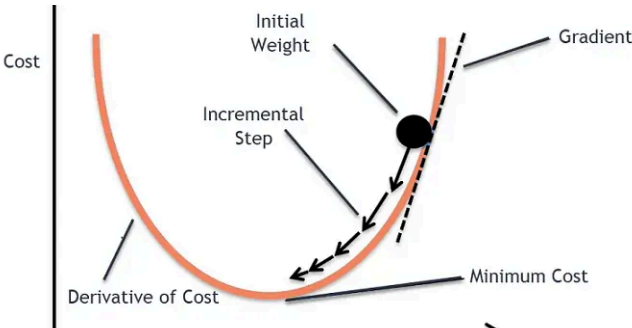
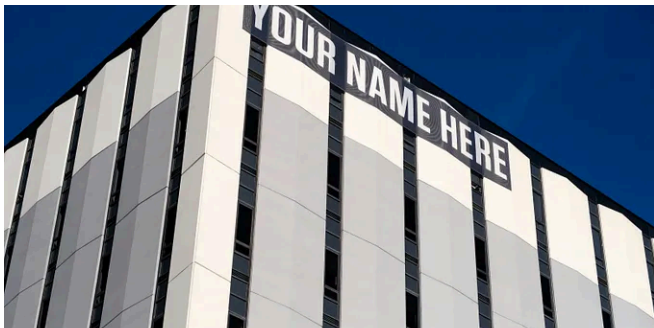
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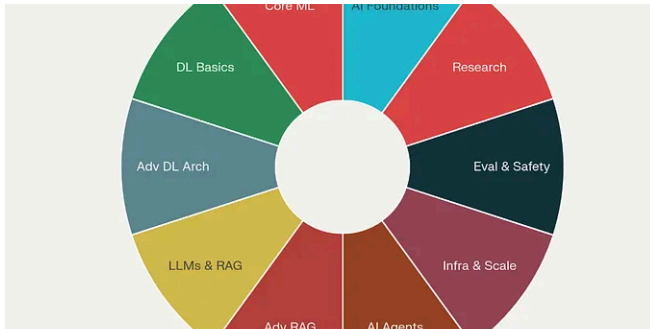
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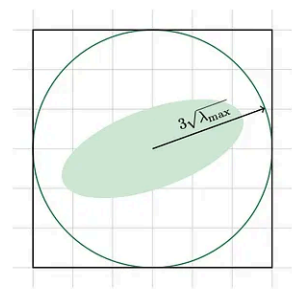
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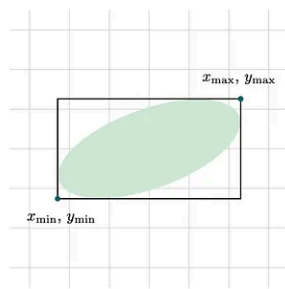
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(a) Conservative Radius



(b) Axis-Aligned SnugBox



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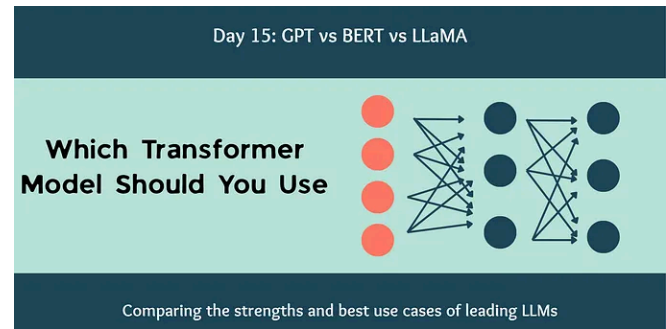
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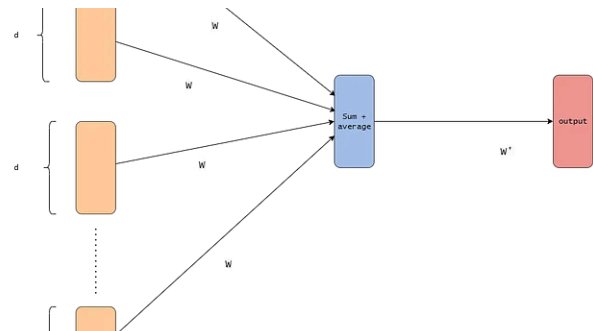
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